

The KOMPOT Code (Johnstone et al. 2018, 2019, 2021; Robeling et al. 2026) versus EXHALE: model comparison and implementation-path analysis

1. Purpose and summary

This memo compares the KOMPOT Code, as documented across its four defining papers, with EXHALE, and then assesses two possible construction paths: writing a KOMPOT-class code from scratch using the published papers, and extending EXHALE with selected KOMPOT physics. The headline verdicts, stated first, are the following.

1. **The two codes appear largely complementary rather than competing.** KOMPOT targets the thermal and chemical structure of weakly to moderately irradiated thermospheres — terrestrial secondary atmospheres, and in the 2026 paper a hydrostatic H_2 giant — while EXHALE targets strongly irradiated transonic H/He + metal winds and the spectral line diagnostics that trace them. The two overlap only in the transonic corner, which is the subject of Johnstone et al. (2019) and the core regime of EXHALE.
2. **The deepest architectural difference is the treatment of chemistry.** KOMPOT integrates a time-dependent kinetic reaction network (Rosenbrock solver) and transports every species by advection and diffusion; EXHALE solves local ionization equilibrium in each cell at every step. EXHALE's choice appears valid where the ionization and recombination timescales are short compared with the flow time, which is its own regime, but it cannot by construction represent transport-limited neutral photochemistry such as hydrocarbon or CO_2 chemistry. That limitation is presumably why EXHALE couples to VULCAN offline for the lower atmosphere rather than solving that chemistry internally.
3. **From-scratch feasibility differs sharply between the papers.** Johnstone et al. (2018) is unusually complete: all governing equations, numerical appendices, and a reaction table running to roughly ten pages. A reimplementa-tion in that operating mode appears feasible provided the external data sets are collected. The new numerics introduced in Johnstone et al. (2019) are underspecified but can be replaced by standard methods. Robeling et al. (2026), by contrast, does not appear reimplementable from the paper alone: the chemical network is not tab-ulated, the CH_4 cooling fit is unpublished, the H_3^+ non-LTE factor is not stated, and the infrared heating channel is never described.
4. **For EXHALE, a few ports look realistic and one does not.** Thermal conduction, the closed-form $\text{CO}_2/\text{NO}/\text{O}$ cooling functions, and a KOMPOT-style subsonic (Jeans-boundary) solution mode are all plausible additions. Adopting kinetic chemistry with species transport would, in contrast, touch the solver core deeply enough that it would amount to writing a new code rather than extending the existing one.

2. The KOMPOT lineage and its three operating modes

The four papers do not describe one fixed code. They describe a common physics core that is run in three distinct hydrodynamic modes: a semi-static mode with a moving exobase (2018, 2021), a fully transonic Godunov mode (2019), and a hydrostatic mode (2026). Table 1 summarizes the lineage; the paragraphs that follow give the detail.

Table 1: The four KOMPOT papers, their operating modes, and their targets.

Reference	Operating mode	Composition and target	Escape regime	Key result
Johnstone et al. (2018), A&A 617, A107	Semi-static, three temperatures, moving exobase	Full N/O/C/H network, 63 species; modern Earth and Venus	Jeans escape at the exobase	Foundation paper; validated against NRLMSISE-00, IRI-2007 and Hedin (1983); solar XUV evolution from -3 to $+2.5$ Gyr
Johnstone et al. (2019), A&A 624, L10	Transonic, MUSCL-Hancock, single temperature	Earth composition (N_2/O_2) at 1 AU	Fully transonic hydrodynamic wind	Earth’s atmosphere under the young active Sun escapes hydrodynamically, $\dot{M} = 1.8 \times 10^9 \text{ g s}^{-1}$
Johnstone et al. (2021), EPSL 576, 117197	Semi-static (2018 mode restored)	Reduced C/N/O network, 30 species; Archean Earth CO_2/N_2	Jeans escape, with a rapid-escape threshold	At least $\sim 40\%$ CO_2 needed at 3.8 Ga; the young Sun was likely a slow rotator
Robeling et al. (2026), arXiv:2608.04086	Hydrostatic ($v = 0$ imposed)	H_2 -rich with hydrocarbons to C_2H_6 ; Jupiter	None (bound atmosphere)	Reproduces the Galileo/JUNO thermal inversion; Joule heating dominates the energy budget as prescribed

2.1 Johnstone et al. (2018): the foundation

The 2018 paper defines the physics core. It is a one-dimensional, time-dependent model carrying three separate temperatures — neutral, ion, and electron — coupled by collisional energy exchange. The hydrodynamics follow the semi-static reduction of Tian et al. (2008): mass and momentum are assumed steady, the velocity ordinary differential equation is integrated downward from the exobase given an imposed exobase outflow speed taken from the Jeans-consistent expressions of Luger et al. (2015), and the density equation is integrated upward from a fixed lower boundary. The upper boundary is the exobase itself and moves with the solution; it is placed where the mean free path equals the pressure scale height, using a constant collision cross section of $2 \times 10^{-15} \text{ cm}^2$.

The chemistry is a kinetic network of 63 species, 30 of them ions, connected by 503 reactions including 56 photoreactions and 7 photoelectron reactions. It is integrated with the RODAS3 Rosenbrock method using an analytic Jacobian, and the chemistry and radiation updates are applied every 100 hydrodynamic steps rather than every step. The model was validated against NRLMSISE-00 and IRI-2007 for the modern Earth and against Hedin et al. (1983) for Venus. Two applications follow: an experiment varying the CO_2 abundance from $0.01\times$ to $1000\times$ the present value, and an evolutionary sequence in solar XUV from 3 Gyr in the past to 2.5 Gyr into the future. In the latter, the exobase neutral temperature at 3 Gyr ago is about 4600 K (with the electron temperature about 5100 K), against a present-day value near 1100 K read from their figure.

The design principle worth flagging is that *no heating efficiency parameter appears anywhere*. The heating rate is instead built up term by term: excess energy from photodissociation, a photoelectron degradation spectrum following Schunk & Nagy (1978) with 100 logarithmic bins between 1 and 1000 eV and local deposition, tabulated reaction

enthalpies, infrared absorption by CO₂ and H₂O using HITRAN/HITEMP line data processed through kspectrum, and a Joule term calibrated to a global total of 1.4×10^{18} erg s⁻¹ for the Earth with a uniform $B = 0.5$ G.

2.2 Johnstone et al. (2019): the transonic mode

The 2019 letter keeps the physics and replaces the hydrodynamics. The semi-static reduction gives way to a MUSCL-Hancock scheme on primitive variables following Toro (1999), and the three temperatures collapse to one because the chosen Riemann solver requires a single temperature. The grid is fixed, spanning altitudes from 50 to 50 000 km in 500 cells, with a zero-gradient outflow condition at the top. Notably, no sonic-point condition is imposed anywhere: the transonic solution emerges from time relaxation.

One model is presented. An Earth-composition (N₂/O₂) atmosphere is placed at 1 AU under the young active Sun, using the Claire et al. (2012) spectrum for 4.4 Gyr ago, in which the flux below 100 nm is enhanced by roughly a factor of 60. The result is a fully transonic wind with $\dot{M} = 1.8 \times 10^9$ g s⁻¹, which would remove the Earth’s present atmosphere in about 0.1 Myr. At the top of the domain the temperature reaches 19 000 K and the outflow speed 7 km s⁻¹, against an escape velocity of 4 km s⁻¹. The outflow is dominated by atomic N and O and reaches up to 20% ionization, and hydrogen is lost about 2.5 times more efficiently than most other species.

2.3 Johnstone et al. (2021): the Archean application

The 2021 paper returns to the 2018 semi-static three-temperature mode; the 2019 numerics are not used. The chemistry is reduced to the C, N and O elements, leaving 30 species with 11 ions and 241 reactions including 42 photoreactions. The lower boundary is a pure CO₂ + N₂ mixture at $T = 267$ K and $n = 5 \times 10^{16}$ cm⁻³ at 50 km altitude. The parameter study crosses base CO₂ mixing ratios from 0.001 to 0.99 with solar activity levels $F_X \approx 2\text{--}15$ erg s⁻¹ cm⁻² at 1 AU (roughly 3–23 times the modern mean) taken along the rotation tracks of Tu et al. (2015).

Escape is counted as Jeans escape at the exobase only. A “rapid escape” regime is diagnosed from a runaway in the exobase solution together with a mass-loss threshold $\dot{M} > 0.1$ bar Myr⁻¹, about 1.7×10^7 g s⁻¹. The principal conclusions are that at 3.8 Ga the atmosphere required at least about 40% CO₂ to survive, that above $F_X \approx 13$ erg s⁻¹ cm⁻² even a pure CO₂ atmosphere escapes rapidly, and that the young Sun must therefore have been a slow rotator.

2.4 Robeling et al. (2026): the giant-planet extension

The 2026 paper extends the code to a giant planet and drops the hydrodynamics entirely: $v = 0$ is imposed in the momentum relation, so the model is hydrostatic, and the exobase is located where the Knudsen number reaches unity. The target is an H₂-rich Jupiter benchmarked against Galileo and JUNO thermal structure and hydrocarbon profiles. The chemistry is extended to hydrocarbons up to C₂H₆, drawn from KIDA, PHIDRATES and several further networks, but the resulting network is not tabulated in the paper. H₃⁺ cooling is taken from Table 5 of Miller et al. (2013) multiplied by a non-LTE scale factor whose value is not stated; CH₄ cooling comes from an LTE fit to ExoMol data over 1–1500 K, with the fit itself deferred to a paper in preparation.

Joule heating dominates the model energy budget and is a scaled free parameter. The total $\Gamma_j \approx 10^{21}$ erg s⁻¹ is taken from Müller-Wodarg et al. (2025) and distributed over a shape function computed from the Pedersen conductivity

with a placeholder unit electric field and a single scalar $B = 4.17$ G. The eddy diffusion coefficient K_{zz} is the other major free parameter, explored using the Moses et al. (2005) profiles A and C together with constant values of 10^6 and 10^7 cm² s⁻¹. The model reproduces the thermal inversion and the upper-atmosphere temperatures well, overpredicts C₂H₂ by a factor of about 2–3, and the authors acknowledge that aerosols, particle precipitation and non-LTE CH₄ are missing. A coupling to TAUREX produces gas-only transmission spectra over 1–15 μm, in which CH₄ dominates from 1 to 10 μm and C₂H₂ beyond 10 μm.

Two internal tensions are worth recording, tentatively. First, the headline that Joule heating dominates appears to follow from the input choice of Γ_j rather than from an independent derivation, since the total is prescribed rather than computed. Second, Table 1 quotes $\Gamma_j \approx 10^{21}$ erg s⁻¹, which is 100 TW, while the discussion of Fig. 3 reports the integrated Joule contribution as about 200 TW; the two statements do not appear reconcilable from the text alone.

3. Physics comparison with EXHALE

The verdict first: within EXHALE’s own regime — strongly irradiated escaping H/He + metal atmospheres and the line diagnostics that probe them — nothing in the four KOMPOT papers supersedes what EXHALE already does, and in several places (Ly α transfer, excited hydrogen, metal line cooling, transit spectra) EXHALE goes considerably further. Conversely, in KOMPOT’s regime — bound or weakly escaping thermospheres with neutral molecular chemistry — EXHALE currently cannot operate at all, because it has no bound-atmosphere solution mode, no thermal conduction, and no kinetic chemistry. Table 2 sets the two side by side. The EXHALE entries were checked against the source tree rather than against its documentation.

Table 2: Physics comparison. KOMPOT entries are as described in the four papers; EXHALE entries were verified in EXHALE/src/.

Aspect	KOMPOT (2018–2026)	EXHALE
Hydrodynamics and regime	Three modes: semi-static with moving exobase (2018, 2021), transonic MUSCL-Hancock (2019), hydrostatic $v = 0$ (2026). Bound and escaping atmospheres both representable.	Single-fluid three-equation Euler system, spherical, Roche potential on by default. Transonic wind only; a bound hydrostatic solution regime does not exist in the code.
Steady-state method	Time relaxation in all modes; convergence criterion not stated in any of the papers.	Explicit SSP-RK3 marching, optionally finished by a pseudo-transient-continuation Jacobian-free Newton–Krylov solver with GMRES (steady_newton.f90). Convergence on the mass-flux spread $du = \text{spread}(\rho v r^2) < 10^{-3}$.
Numerical scheme	2019: MUSCL-Hancock on primitive variables, Riemann solver cited only by equation number in Toro (1999). 2018/2021: ODE integration, no Riemann solver.	PLM or WENO3 reconstruction; HLLC, Roe or LLF fluxes (Num_Fluxes.f90, Reconstruction.f90).
Temperatures	Three (T_n , T_i , T_e) with collisional exchange in 2018, 2021, 2026; collapsed to one in 2019 because the Riemann solver requires it.	One temperature shared by neutrals, ions and electrons, $T = p / [(n_{\text{tot}} + n_e)k]$. No T_e anywhere in the source.
Thermal conduction	Present in all modes, with separate conductivities for neutrals, ions and electrons; dominant near the base of a bound thermosphere.	Absent. The conduction slot in the energy source vector is identically zero (Source.f90, S(3) = 0.0); the only mentions of conduction in the source are comments recording that it is deferred.
Photo-heating and heating efficiency	Built from first principles; no heating efficiency parameter anywhere, stated explicitly as a design goal.	Photoionization heating computed from the frequency integral, also with no efficiency parameter. The η reported in the output is a diagnostic, not an input. The design stance is the same in both codes.

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Table 2, continued

Aspect	KOMPOT (2018–2026)	EXHALE
Secondary and photoelectron treatment	Photoelectron degradation spectrum transported over 100 logarithmic bins from 1 to 1000 eV (Schunk & Nagy 1978), with local deposition.	Secondary ionization by fast photoelectrons through the Shull & van Steenberg (1985) fits: a local energy-partition treatment, not a degradation-spectrum calculation.
Chemical heating	General bookkeeping of tabulated reaction enthalpies across the whole network.	Only specific channels: He(2^3S) + H and He(2^3S) + H ₂ Penning ionization (6.2 and 4.4 eV), He recombination-radiation coupling, and photoelectrons from H($n = 2$) Balmer-continuum ionization. No general reaction-enthalpy bookkeeping.
Joule heating	Present. Earth: total 1.4×10^{18} erg s ⁻¹ , $B = 0.5$ G. Jupiter: total $\Gamma_j \approx 10^{21}$ erg s ⁻¹ distributed over a Pedersen conductivity shape, $B = 4.17$ G.	Absent. A search of the source for Joule, magnetic or Pedersen terms returns no hits; there is no magnetic field in the model.
Infrared and bolometric heating	CO ₂ and H ₂ O infrared absorption (2018) from HITRAN/HITEMP; an infrared heating channel is referred to but not described in 2026.	No bolometric infrared heating in the production solve. The energy grid floor is 4.8 eV (or the lowest metal ionization potential); the only sub-XUV stellar inputs are a diluted-blackbody Balmer continuum acting on H($n = 2$) and stellar Ly α pumping.
Cooling channels	CO ₂ 15 μ m cool-to-space (two-level, Kumer & James escape probabilities), NO 5.3 μ m, O 63/147 μ m, H ₂ O; plus H ₃ ⁺ and CH ₄ in 2026.	Ly α ; H and He recombination, collisional ionization and free-free; He 2^3S channels; metal lines as closed-form CHIANTI v11 fits for C I/II, N I/II, O I/II, Mg I/II, Ca II, Na I, Fe I/II, with density-dependent [C II] 158 μ m and [O I] 63 μ m saturation floors and a statistical-equilibrium Fe II coefficient; H ₃ ⁺ infrared cooling (h3p_cooling.f90). No CO ₂ , NO or CH ₄ cooling.
H ₃ ⁺ cooling detail	Miller et al. (2013) Table 5 fits times a non-LTE scale factor whose value is not given (2026).	The same Miller et al. (2013) Table 5 fits, plus the Table 6 non-LTE departure factor. On this one item EXHALE appears better documented than Robeling et al. (2026).
Chemistry treatment	Time-dependent kinetic network integrated with RODAS3 Rosenbrock and an analytic Jacobian; every species carries its own continuity equation.	Local steady-state ionization equilibrium solved in every cell at every step (util_ion_eq.f90, ionization_equilibrium.f90), by an analytic-Jacobian Newton method with a MINPACK hybrd1 fallback.
Species set	2018: 63 species, 30 ions, 503 reactions. 2021: 30 species, 11 ions, 241 reactions. 2026: extended to hydrocarbons up to C ₂ H ₆ (network not tabulated).	37 species: H I/II; He I/II/III and He 2^3S ; 27 metal ion stages of the ten elements C, N, O, Mg, Si, Ca, Na, K, S, Fe (species_table.f90); and H ₂ , H ₂ ⁺ , H ₃ ⁺ , HeH ⁺ . Charge exchange from Huang et al. (2023) Table 4; the molecular network is 23 reactions with Koskinen et al. (2022) rates, also in local equilibrium. H ₂ Lyman–Werner photodissociation is absent, as noted in the code comments.
Species transport	Every species is advected and diffused; molecular diffusion coefficients from Banks & Kockarts with an N ₂ background, plus eddy diffusion.	No species advection in the main loop; an advection-corrected re-solve exists only as a post-processing step (post_process_adv.f90). He/H element-ratio diffusion with binary molecular D , eddy K_{zz} , ambipolar-corrected settling (on by default within the option) and optional thermal diffusion (species_diffusion.f90); each metal may diffuse independently. The whole diffusion option is off by default, and there is no eddy diffusion of the bulk gas.
Radiative-transfer geometry	Single slant ray at a chosen zenith angle (66° for Earth, 60° in 2026); optical depth summed over all photoreactions.	Radial column with exp($-\tau$) attenuation and a day–night redistribution factor ξ (0.5 or 0.25). Spectrum from a broken power law, a tabulated SED, or monochromatic; default range 13.6–1240 eV (the floor drops to 4.8 eV with He 2^3S , or to the lowest neutral-metal ionization potential), in about 200 bins. The two treatments are different one-dimensional averaging approximations, not a right-versus-wrong distinction.

Table 2, continued

Aspect	KOMPOT (2018–2026)	EXHALE
Ly α transfer and excited hydrogen	Not treated.	Ly α radiative transfer through a Neufeld escape-probability closure computed in line, or an imported mean-intensity profile from the LaRT Monte Carlo code (<code>lya_rt.f90</code>); full non-LTE H($n=2$) 2s/2p balance feeding Balmer-continuum ionization and heating (<code>excited_hydrogen.f90</code>). Nothing comparable exists in KOMPOT.
Boundaries and domain	Lower boundary fixed at 50 km (2019, 2021) or 65 km (2018 Earth). Upper boundary at the moving exobase (2018, 2021, 2026) or zero-gradient outflow at 50 000 km (2019).	500 cells by default from the base radius to the Hill or L1 radius set by the Roche geometry (overridable). Fixed base density and temperature, with T_0 the equilibrium temperature, plus a one-way valve enforcing $v \geq 0$ at the base; zero-gradient outflow at the top; $\dot{M}$ read near the top.
Free parameters	Eddy K_{zz} , described by the authors as the only real free parameter (2019); the Joule total and its electric field; the H $_3^+$ non-LTE factor (2026).	Eddy K_{zz} when diffusion is enabled, and the day–night redistribution factor ξ .
Outputs and observables	Thermal, chemical and ionization structure; exobase state; Jeans or hydrodynamic mass-loss rate; gas-only transmission spectra through TAUREX in 2026.	Thermal, chemical and ionization structure; $\dot{M}$; and transit transmission spectra for He I 10830, Ly α , H α , H β , Mg II h&k, Ca II H&K and Na I D.

3.1 Free parameters, stated honestly

The KOMPOT papers make a point of using no heating efficiency parameter, and that claim appears accurate. It should not be read, however, as meaning that the models are free of tuned inputs. By the authors’ own statement the eddy diffusion coefficient is a free parameter; the Joule heating total and the electric field used to shape its deposition are prescribed rather than derived; and the H $_3^+$ non-LTE scale factor in the 2026 paper is an unstated multiplier. EXHALE shares the first of these (K_{zz} , when diffusion is switched on) and adds the day–night redistribution factor ξ , which plays a role loosely analogous to KOMPOT’s choice of zenith angle. On the specific question of heating efficiency the two codes take the same first-principles position.

4. Reproducibility of each paper

Verdict first: Johnstone et al. (2018) comes close to a complete specification of a working code; the 2019 and 2021 papers inherit that specification but leave their own new pieces underdocumented; and Robeling et al. (2026) does not appear reproducible from the paper alone.

4.1 Johnstone et al. (2018)

What is fully specified is unusually broad. It includes all governing equations, including the semi-static reduction and the directions in which each ODE is integrated; the radiation scheme (10–4000 Å in 1000 bins, infrared 1–20 μm in roughly 10^4 selected bins, ray step $H/5$, optical depth summed over photoreactions); the photoelectron degradation equation with its bin structure and all cross-section sources; every heating and cooling formula together with its constants (the CO $_2$ cool-to-space two-level model with Kumer & James escape probabilities and the collider rate table, NO 5.3 μm , O 63 and 147 μm , and the conductivities for neutrals, ions and electrons); the diffusion coefficients and thermal-diffusion factors; all numerical solvers at appendix level of detail (Crank–Nicolson tridiagonal forms,

RODAS3 with tolerances, MUSCL/TVD parameters); the complete reaction table with rates, enthalpies and validity ranges (Table H.1); the operator-splitting order and the 100-step update cadence; and a set of solver validations against an analytic Parker wind, a Sod tube compared with VAC, a cross-check against KROME, and a photoelectron spectrum compared with Singhal & Haider (1984).

What is left out is a shorter but not empty list: the number of grid cells and the exact spacing law; the steady-state convergence criterion, which is likely the largest single obstacle to reproducing the quoted numbers; the Courant number, given only as $C \sim 1$; the Joule electric field, which is tuned to a global total and is therefore not reconstructible for a single column; the H_2O cooling equations, for which the reader is pointed to Kasting & Pollack (1983); the algorithm for the thermal/non-thermal crossover energy E_t ; the energy-loss tabulations ΔE ; the ion and electron cooling terms $Q_{c,i}$ and $Q_{c,e}$, which appear in the equations but are never given; and the exobase regridding algorithm. Beyond the paper, an implementer would need the external numerical arrays: PHIDRATES cross sections; electron-impact cross sections from Green & Barth (1965), Watson et al. (1967), Jackman et al. (1977) and Jusick et al. (1967); the Banks & Kockarts and Schunk & Nagy collision tables; HITRAN/HITEMP line lists processed through kspectrum; and the Claire et al. (2012) spectra.

4.2 Johnstone et al. (2019)

The physics is inherited from 2018 and is therefore documented. What the letter adds is a change of numerics, and that change is underspecified: the Riemann solver is identified only by an equation number in Toro (1999); the limiter, the operator-splitting order, the CFL number, the sonic-point cutoff used to disable diffusion, the definition used for the mass-loss rate, and the convergence criterion are all absent. The 50 km base state — temperature, density, and all 63 mixing ratios — is not tabulated, and it cannot simply be copied from the 2018 paper because that Earth model was based at 65 km. None of these gaps involves physics, so all of them can be closed by an implementer's own reasonable choices, at the cost of not reproducing the quoted numbers exactly.

4.3 Johnstone et al. (2021)

This paper is largely derivable from 2018 by mechanical means: the element filter rule is given, and the CO_2/N_2 base state is in fact better documented than the 2019 base state. The real gaps are the eddy diffusion coefficient adopted for a CO_2 -rich atmosphere, which directly controls how much CO_2 reaches the radiating region and therefore controls the headline threshold; whether Joule heating was enabled; the grid; the set of species allowed to escape by Jeans escape; and the conversion used between bar Myr^{-1} and g s^{-1} .

4.4 Robeling et al. (2026)

Four items block an independent implementation, and for each of them there is no citable closing source. The chemical network is not tabulated, and it is described as the union of six cited sources with an undocumented selection rule. The CH_4 cooling fit is deferred to a paper in preparation. The H_3^+ non-LTE scale factor is given neither a value nor a reference. The infrared heating channel appears in the abstract and in a figure legend but is never described. Secondary omissions are the epoch of the solar spectrum and the orbital distance, and the internal 100 TW versus 200 TW

inconsistency noted in §2.4. The conclusion is that an independent implementation could only be built “in the style of” Robeling et al. (2026) — for instance, with a VULCAN-derived network and fits of the implementer’s own construction — and should not be described as a reproduction.

5. Implementation paths

5.1 Building a KOMPOT-class code from scratch

A reimplementaion in the 2018 operating mode appears feasible as a standalone project. The paper together with its appendices amounts to a near-complete specification, and the missing numerical items (grid, convergence criterion, Courant number) are choices any implementer can make without guessing at physics. The labor would be dominated not by the equations but by collecting the external data — cross-section and rate archives, collision tables, line lists — and by entering the 503-reaction network. Reproducing the published curves qualitatively appears realistic on that basis; matching the quoted numbers exactly does not, because the convergence criteria are not published. Adding the transonic mode of 2019 requires an independent choice of Godunov-type numerics, for which EXHALE-grade methods would be entirely adequate. None of this requires inventing new physics.

5.2 Extending EXHALE

Table 3 lists the candidate ports with a verdict, a rough effort estimate, and an assessment of physical value; the notes below expand on the entries that need it.

Table 3: Candidate KOMPOT-inspired additions to EXHALE.

	Candidate	Effort	Verdict and physical value
(a)	CO ₂ /NO/O closed-form cooling (2018 Eqs. 34–43 plus the collider table)	low	Trivial to code, but meaningless without the corresponding neutral species. Valuable only together with (d).
(b)	Thermal conduction	moderate	Realistic port. All coefficients are in the 2018 paper; for a single-temperature gas the density-weighted mixing of 2019 applies. Prerequisite for (c).
(c)	Subsonic/Jeans solution mode	large	The most scientifically interesting port. Would extend EXHALE to weakly irradiated planets where no transonic wind exists.
(d)	Kinetic chemistry with species transport	very large	Not an extension. Would effectively be a new code.
(e)	Schunk & Nagy photoelectron degradation	moderate	Closed specification, but marginal gain in EXHALE’s strongly ionized regime.
(f)	Joule heating in the Robeling form	low	Technically easy, but imports an unconstrained free parameter. Sensitivity studies only.
(g)	Robeling-style Jupiter infrared thermosphere	—	Out of reach as EXHALE stands; needs (d) and (b) plus a hydrostatic mode.

(b) Thermal conduction. The effort is moderate rather than small because the term must enter in two places: EXHALE already has an implicit energy-update pattern that can be extended, but the Jacobian-free Newton–Krylov residual would also need the conduction term for the steady solver to remain consistent. Physically, conduction is negligible where advection dominates, which is the case in EXHALE’s converged winds, but it matters near the base and it matters throughout for weakly irradiated planets. That is why it is a prerequisite for (c) rather than an end in itself.

(c) A subsonic/Jeans solution mode. This means a semi-static hydrodynamic solve with a moving exobase and a Jeans-consistent upper boundary, and it is structurally the largest realistic addition on the list. The algorithm is documented completely in Appendix D of the 2018 paper, so the specification risk is low even though the implementation effort is high. The scientific argument for it is that EXHALE currently cannot represent a bound thermosphere at all: the base valve and the outflow top together assume a wind. Adding this mode would extend the code’s planet class downward, to planets where a transonic wind does not exist.

(d) Kinetic chemistry with species transport. This is the chasm in the comparison. Replacing local ionization equilibrium with a Rosenbrock network integration plus advection and diffusion of every species would touch the solver core, the Newton–Krylov residual, and the post-processing chain simultaneously. It is better described as writing a new code than as extending EXHALE. If terrestrial-composition escape — Archean-Earth type questions — ever becomes a research goal here, a fresh implementation guided by the 2018 paper appears more sensible than bending EXHALE to fit.

(f) and (g). Joule heating is easy to add in the Robeling form, since it amounts to distributing a prescribed total over a Pedersen conductivity shape, but for exoplanets both B and the effective electric field are unknown, so its value appears limited to sensitivity studies. As for a Robeling-style Jupiter application, it is out of reach for EXHALE as it stands. It is worth noting, though, that EXHALE already has the H_3^+ cooling piece, and that it reaches the neutral molecular lower atmosphere by a different route — the offline VULCAN coupling — rather than by solving that chemistry internally.

5.3 Recommendation

Tentatively, the comparison suggests keeping the two codes’ regimes separate rather than trying to make either one cover both. For EXHALE’s own science, (b) and (c) appear to be the natural KOMPOT-inspired upgrades: they extend the range of planets the code can address without disturbing what it already does well. Items (a) and (d) are worth revisiting only if secondary atmospheres become a research goal, and in that case a new code guided by the 2018 paper looks more sensible than an extension; item (f) is worth having only as a sensitivity knob. Finally, the KOMPOT papers independently validate EXHALE’s design stance on first-principles heating: both codes compute heating term by term, neither uses a heating efficiency parameter, and the KOMPOT authors treat that as a selling point.

6. Caveats

This memo rests on the four papers — published text for the three Johnstone papers, preprint text for Robeling et al. (2026) — together with the EXHALE source tree. The KOMPOT source is not public, so every statement here about KOMPOT is a statement about what its papers say, and the code may contain physics or numerics they do not describe. Several numbers above are read from published figures rather than tabulated values: some Johnstone et al. (2021) thresholds, the present-day exobase temperature in Johnstone et al. (2018), and the Robeling et al. (2026) temperatures. The effort estimates in §5.2 are qualitative. The EXHALE statements were checked in the source at the time of writing and describe the default configuration; several features discussed are opt-in.